

M003 Week 8 — English Worksheet

Topic: Final — Two-Faced Statue · **Course:** 3D Coordinates · **Time:** about 45 minutes

This is your final build. No new commands — just everything from this course working together. Your statue shows **one picture from the front** and a **different picture from the side**. The front face is flat along **x and y**, the side face is flat along **z and y**, and they meet at one corner edge.

1 · Predict 🧠

Read each plan. Before you imagine it happening, write what you think each **view** will show.

```
fill stone from (0, 0, 0) to (4, 4, 0)
```

This wall stretches along x and y, and z stays 0. From the front you see a 5×5 square. What do you see from the side — the same square or something much thinner?


```
front face: fill stone from (0, 0, 0) to (4, 4, 0)
side face:  fill stone from (0, 0, 0) to (0, 4, 4)
```

Two faces, both 5 tall. Do they touch each other? Which blocks belong to both faces at the same time?


```
front face: fill stone from (0, 0, 0) to (4, 4, 0)
side face:  fill stone from (0, 0, 0) to (0, 2, 4)
```

Now the side face only goes up to height 2. Standing at the corner, do the two faces look like one statue or like a mistake? Why?

2 · Spot the Bug 🐛

Each block of code below was meant to do something, but it is broken. Read what the code is **supposed** to do, then rewrite it so it works. After that, explain why the original was wrong and why your fix works.

Bug A — The two faces should meet only at the **corner edge** at $x = 0, z = 0$. Right now the side face slices through the **middle** of the front picture, so the front view is ruined.

```
fill stone from (0, 0, 0) to (4, 4, 0)
fill stone from (2, 0, 0) to (2, 4, 4)
```

Hint: look at the **x** numbers in the second line.

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug B — Both faces should be the **same height** — up to $y = 4$. Right now one face is shorter than the other.

```
fill stone from (0, 0, 0) to (4, 4, 0)
fill stone from (0, 0, 0) to (0, 2, 4)
```

Write the fixed code:

Why was it wrong? Why does your fix work?

Bug C — Both faces should sit on the **ground**, starting at $y = 0$. Right now one face floats one block in the air.

```
fill stone from (0, 0, 0) to (4, 4, 0)
fill stone from (0, 1, 0) to (0, 5, 4)
```

Hint: compare the **y** numbers where each face starts.

Write the fixed code:

Why was it wrong? Why does your fix work?

3 · Tell Me What You Built 📷

Plan **before** you build. Sketch your **front view** picture as a grid (use # for stone, . for air):

Sketch your **side view** picture as a grid — same height as the front:

List the corner coordinates of each face — front face from (... , ... , ...) to (... , ... , ...), side face from (... , ... , ...) to (... , ... , ...):

Now switch to your homework world and build your two-faced statue. Check your build against this list:

- ☐ The front view shows one clear picture.

- ☐ The side view shows a **different** clear picture.
- ☐ Both faces are the **same height**.
- ☐ Both faces start at the **ground** — the bases line up.
- ☐ The faces meet at one corner edge, not through the middle.

Send a photo or video of **both views**, then explain what you did. Use these sentence starters — write 4 to 6 sentences total.

My front picture is ... and my side picture is ...

The two faces meet at ...

I made both faces the same height by ...

One tricky moment was when ...

If I had more time, I would ...

The two faces meet at ...

I made both faces the same height by ...

One tricky moment was when ...

If I had more time, I would ...

4 · Record Your Walkthrough

Now take a video on your phone (or a parent's phone) while you walk around your statue. Talk through it like you are teaching someone who has never seen it. Try to use these words: **front view**, **side view**, **face**, **corner**, **height**.

1. Stand in front of the statue and say what the front picture shows.
2. Walk around to the side and say what the side picture shows.
3. Point at the corner edge where the two faces meet and say its coordinates.
4. Say in your own words how planning with coordinates made this build possible.

Write what you will say in your video. Use the space below to plan it before you record — you can read from it while filming.

Submit 

Send this worksheet + your walkthrough video to teacher on KakaoTalk.